

## MORIOKA, Japan

WMO: 475840

Lat: 39.6986N

Lon: 141.1658E

Elev: 510

StdP: 14.43

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 16.5       | 19.1 | 8.6                           | 8.7 | 22.7 | 10.6 | 9.6 | 24.4 | 18.9                  | 31.7 | 16.7 | 31.6 | 4.3                   | 130  | 0.482 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 14.3                   | 89.0            | 75.2 | 86.2 | 73.7 | 83.5 | 72.3 | 76.8                | 85.4 | 75.6 | 83.0 | 74.5 | 81.0 | 9.2                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 74.5                            | 131.7 | 80.2 | 73.5 | 126.9 | 79.5 | 72.4 | 122.3 | 78.7 | 40.7          | 85.7 | 39.6 | 83.2 | 38.5 | 81.0 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 18.4              | 16.1 | 14.3 | 12.0                       | 93.3 | 3.0                | 2.5 | 9.8                                                | 95.1 | 8.0        | 96.5 | 6.3        | 97.9 | 4.1        | 99.7 |    |    |
|                   |      |      | 11.0                       | 78.9 | 2.9                | 1.3 | 8.9                                                | 79.9 | 7.2        | 80.6 | 5.6        | 81.4 | 3.5        | 82.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 51.5   | 28.8  | 30.2  | 36.8  | 47.7  | 58.5  | 66.2  | 73.4  | 75.4  | 67.5  | 55.3  | 43.3  | 33.6  |
|                                                                   | DBStd       | 17.15  | 4.59  | 4.97  | 5.66  | 6.50  | 5.67  | 4.85  | 4.96  | 5.29  | 6.07  | 5.67  | 6.05  | 5.05  |
|                                                                   | HDD50       | 2485   | 658   | 553   | 412   | 119   | 6     | 0     | 0     | 0     | 16    | 214   | 507   |       |
|                                                                   | HDD65       | 5723   | 1123  | 973   | 876   | 519   | 213   | 44    | 3     | 2     | 42    | 305   | 651   | 972   |
|                                                                   | CDD50       | 3036   | 0     | 0     | 1     | 50    | 268   | 486   | 726   | 788   | 525   | 179   | 13    | 0     |
|                                                                   | CDD65       | 797    | 0     | 0     | 0     | 0     | 10    | 79    | 264   | 325   | 117   | 2     | 0     | 0     |
|                                                                   | CDH74       | 5272   | 0     | 0     | 0     | 9     | 141   | 462   | 1705  | 2366  | 585   | 4     | 0     | 0     |
|                                                                   | CDH80       | 1489   | 0     | 0     | 0     | 0     | 22    | 79    | 498   | 778   | 112   | 0     | 0     | 0     |
|                                                                   | WSAvg       | 6.4    | 6.0   | 6.5   | 7.3   | 7.7   | 7.5   | 6.8   | 6.3   | 6.2   | 5.8   | 5.6   | 5.9   | 5.9   |
|                                                                   | PrecAvg     | 52.0   | 2.1   | 2.0   | 3.2   | 3.6   | 4.1   | 4.6   | 7.6   | 7.2   | 6.4   | 4.4   | 3.8   | 3.0   |
| Precipitation                                                     | PrecMax     | 67.6   | 4.5   | 5.0   | 5.3   | 7.3   | 7.6   | 8.1   | 17.9  | 17.2  | 11.5  | 9.8   | 7.3   | 9.6   |
|                                                                   | PrecMin     | 35.2   | 0.9   | 0.8   | 1.1   | 0.9   | 1.5   | 1.8   | 1.9   | 2.0   | 2.0   | 0.7   | 0.9   | 1.0   |
|                                                                   | PrecStd     | 8.3    | 1.0   | 0.9   | 1.3   | 1.7   | 1.5   | 1.7   | 3.5   | 4.0   | 2.7   | 2.0   | 1.4   | 1.5   |
|                                                                   |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 43.5  | 48.5  | 61.6  | 74.6  | 83.1  | 85.8  | 91.4  | 93.5  | 87.3  | 73.8  | 63.2  | 52.0  |
|                                                                   |             | MCWB   | 38.5  | 40.9  | 48.1  | 54.8  | 62.7  | 67.8  | 75.7  | 76.9  | 73.9  | 62.6  | 54.5  | 47.2  |
|                                                                   | 2%          | DB     | 39.8  | 44.2  | 54.8  | 68.9  | 77.8  | 82.3  | 88.2  | 90.2  | 83.3  | 70.2  | 59.4  | 47.9  |
|                                                                   |             | MCWB   | 35.0  | 38.1  | 45.6  | 53.4  | 60.0  | 66.6  | 74.7  | 76.4  | 72.5  | 60.7  | 52.4  | 43.5  |
|                                                                   | 5%          | DB     | 37.7  | 41.0  | 50.6  | 64.2  | 73.9  | 79.2  | 85.4  | 87.2  | 79.9  | 67.9  | 56.5  | 44.5  |
|                                                                   |             | MCWB   | 33.6  | 35.6  | 43.1  | 51.5  | 59.0  | 65.7  | 73.4  | 74.9  | 70.2  | 60.1  | 50.8  | 40.3  |
|                                                                   | 10%         | DB     | 35.9  | 38.5  | 46.8  | 60.1  | 70.1  | 76.2  | 82.5  | 84.4  | 77.2  | 65.3  | 53.8  | 41.7  |
|                                                                   |             | MCWB   | 32.2  | 33.7  | 40.2  | 49.5  | 57.4  | 64.6  | 72.4  | 73.9  | 69.2  | 58.5  | 49.2  | 37.6  |
|                                                                   | 0.4%        | WB     | 39.5  | 42.8  | 50.9  | 60.5  | 65.7  | 72.0  | 78.1  | 78.8  | 76.6  | 67.4  | 57.4  | 48.1  |
|                                                                   |             | MCDB   | 42.2  | 46.4  | 57.7  | 66.5  | 76.2  | 78.7  | 87.4  | 89.7  | 82.6  | 69.5  | 60.5  | 50.9  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 36.3  | 39.3  | 47.0  | 56.0  | 63.4  | 69.7  | 76.3  | 77.3  | 74.5  | 64.0  | 54.5  | 44.4  |
|                                                                   |             | MCDB   | 38.4  | 42.9  | 52.9  | 64.7  | 71.8  | 76.3  | 84.7  | 86.4  | 80.2  | 67.7  | 57.7  | 46.9  |
|                                                                   | 5%          | WB     | 34.5  | 36.4  | 44.0  | 53.5  | 61.5  | 68.0  | 75.0  | 76.3  | 72.9  | 61.6  | 52.2  | 41.0  |
|                                                                   |             | MCDB   | 36.8  | 40.2  | 49.6  | 60.9  | 69.2  | 75.0  | 82.1  | 84.3  | 78.4  | 65.5  | 55.4  | 43.9  |
|                                                                   | 10%         | WB     | 32.8  | 34.1  | 41.1  | 51.1  | 59.6  | 66.5  | 73.9  | 75.2  | 70.7  | 59.6  | 49.7  | 38.1  |
|                                                                   |             | MCDB   | 35.3  | 37.6  | 46.1  | 57.8  | 67.0  | 73.2  | 80.0  | 82.3  | 75.6  | 63.9  | 53.0  | 41.0  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.6  | 13.1  | 15.1  | 18.5  | 18.6  | 16.3  | 13.9  | 14.3  | 14.9  | 16.7  | 14.9  | 11.2  |
|                                                                   |             | MCDBR  | 12.4  | 15.8  | 21.3  | 26.1  | 26.7  | 22.1  | 18.3  | 17.9  | 16.8  | 18.8  | 18.6  | 14.7  |
|                                                                   |             | MCWBR  | 9.4   | 11.3  | 13.9  | 13.7  | 12.3  | 9.5   | 7.4   | 6.7   | 7.6   | 10.6  | 12.9  | 11.5  |
|                                                                   | 5% WB       | MCDBR  | 11.5  | 14.7  | 19.4  | 19.9  | 19.8  | 16.1  | 15.0  | 15.9  | 14.0  | 14.2  | 16.0  | 14.2  |
|                                                                   |             | MCWBR  | 9.7   | 11.5  | 14.3  | 12.5  | 10.7  | 7.9   | 6.7   | 6.6   | 7.2   | 9.7   | 12.6  | 11.9  |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.374 | 0.421 | 0.501 | 0.543 | 0.556 | 0.537 | 0.542 | 0.518 | 0.456 | 0.423 | 0.403 | 0.381 |
|                                                                   | taud        |        | 1.963 | 1.831 | 1.771 | 1.777 | 1.804 | 1.922 | 1.983 | 2.053 | 2.185 | 2.164 | 2.116 | 2.061 |
|                                                                   | Ebn at Noon |        | 235   | 241   | 235   | 236   | 237   | 241   | 239   | 240   | 249   | 242   | 226   | 223   |
|                                                                   | Edh at Noon |        | 41    | 54    | 64    | 68    | 68    | 60    | 56    | 51    | 42    | 38    | 35    | 35    |
|                                                                   | RadAvg      |        | 545   | 794   | 1093  | 1401  | 1566  | 1543  | 1366  | 1364  | 1139  | 901   | 607   | 457   |
| All-Sky Solar Radiation                                           | RadStd      |        | 39    | 59    | 77    | 133   | 156   | 117   | 171   | 167   | 93    | 78    | 47    | 42    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (16 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page